Calcurion Physics Worksheets

A-Level Physics

Topic: Nuclear Physics

MARKSCHEME

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1. Energy required to separate nucleus into individual nucleons **A2** [2]
2. Stability of the nucleus **A1** [1]
3. Peak near Fe; fission left, fusion right of peak **A3** [3]
4. 1/12 mass of ^{12}C atom **A1** [1]
5. 931 MeV **M1A1** [3]
6. $N = N_0 e^{-\lambda t}$ **A1** [1]
7. Probability of decay per unit time **A1** [1]
8. $T_{1/2} = 69.3$ s **M1A1** [2]
9. $A = 1.0 \times 10^{10}$ Bq **M1A1** [2]
10. Heavy nucleus splits into lighter nuclei **A2** [2]
11. Light nuclei join to form heavier nucleus **A2** [2]
12. High energy density / low CO_2; radioactive waste **A2** [2]
13. $^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^{4}_{2}\text{He}$ **A2** [2]
14. $^{14}_{6}\text{C} \rightarrow {}^{14}_{7}\text{N} + {}^{0}_{-1}\text{e} + \bar{\nu}$ **A2** [2]
15. Natural radiation always present **A1** [1]
16. $BE = 28.3$ MeV **M1A1** [2]
17. 7.07 MeV per nucleon **M1A1** [2]
18. 1/8 **A2** [2]
19. 3 half-lives; $T_{1/2} = 4.0$ h **M1A1** [3]
20. Control rods absorb neutrons; moderator slows neutrons **A3** [3]
21. $N = 2.55 \times 10^{21}$; $E = 8.2 \times 10^{10}$ J **M1A1** [4]
22. High temperature and density (plasma confinement) **A2** [2]
23. Electrostatic repulsion; need high KE to overcome **A3** [3]
24. $R = 2.75$ fm **M1A1** [2]
25. Neutron absorption; unstable; splits releasing neutrons and energy **A3** [3]